			$T = 181 + 273 + 87 = 541$ K	1
5	**(a)**	**(i)**	Given $I = nAve$, since I, n, e are the same for cross-sections X and Y, $$\frac{v_Y}{v_X} = \frac{A_X}{A_Y} = \frac{\pi d_X^2 / 4}{\pi d_Y^2 / 4} = \frac{d_X^2}{d_Y^2}$$ $$= (\frac{d}{0.72d})^2 = 1.93$$	1 1
		(ii)	$$R = \frac{\rho l}{A} = \frac{(1.12 \times 10^{-6})(4 \times 10^{-3})}{\pi(\frac{(0.72 \times 0.21 \times 10^{-3})^2}{4})}$$ $= 0.250\ \Omega$	1 1

		(iii)	When the cross-sectional area is smaller, its resistance per unit length is larger. With the same current through entire wire, the voltage per unit length of the damaged part is larger.	1
	(b)	**(i)**	$I_1 + I_3 = I_2$	1
		(ii)	p.d. across BJ of wire changes so there is a difference between the p.d across wire points BJ and p.d. across cell E	1 1
		(iii)	$V_{\mathrm{BF}} = \dfrac{R_2}{R_1 + R_2 + r_1} \times E_1 = \dfrac{3.5}{3.0 + 3.5 + 1.5} \times 4.5 = 1.96875\ \mathrm{V}$ $\dfrac{l_{BJ}}{l_{BF}} = \dfrac{E_2}{V_{\mathrm{BF}}} \Rightarrow l_{BJ} = \dfrac{E_2}{V_{\mathrm{BF}}} \times l_{BF} = \dfrac{1.2}{1.96875} \times 1.0 = 0.610\ \mathrm{m}$	1 1 1

6	**(a)**	**Faraday's law** of electromagnetic induction states that the **magnitude** of the	